

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is: a Denial of Service (DoS) attack, specifically a **TCP SYN flood attack**, targeting port 443 on the web server (192.0.2.1).

The logs show that: a single source IP address (203.0.113.0) is sending a massive, continuous burst of [SYN] packets to the web server without completing the TCP three-way handshake by returning the final [ACK] packet.

This event could be: a deliberate malicious intrusion designed to exhaust the server's connection backlog table and memory, preventing legitimate users (like 198.51.100.23 and 198.51.100.14) from connecting to the site.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. **SYN (Synchronize):** The client sends a [SYN] packet to the web server requesting to establish a connection.
2. **SYN-ACK (Synchronize-Acknowledge):** The web server acknowledges the request, allocates memory resources in its connection queue, and returns a [SYN, ACK] packet to the client.
3. **ACK (Acknowledge):** The client responds with an [ACK] packet to confirm the connection is active, allowing data transfer (such as HTTP/HTTPS web traffic) to begin.

Explain what happens when a malicious actor sends a large number of SYN packets all at once:

The server acknowledges each incoming [SYN] request by returning a [SYN-ACK] and places each connection into a "half-open" state while waiting for the client's final response. When an attacker transmits hundreds of [SYN] requests simultaneously and intentionally

withholds the final [ACK] packets, the server's connection backlog queue fills up completely. This depletes system memory and socket capacity, leaving no room to process legitimate requests.

Explain what the logs indicate and how that affects the server:

The packet capture log reveals that a single external host (203.0.113.0) generated over 130 rapid [SYN] requests to port 443 on the destination web server (192.0.2.1). As the server's backlog queue became overwhelmed, it began issuing [RST, ACK] (Reset) packets to terminate connections. Consequently, legitimate client machines (such as 198.51.100.23 and 198.51.100.14) experienced severe delays, connection timeouts, and an inability to retrieve web pages like /sales.html